# Risk Aviation Project

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# Summary

- **Objective:** Recommend aircraft types and models based on safety data.
- **Data Sources:** Fatality rates by engine configuration, engine type and aircraft make/model.
- **Scope:** General and commercial aviation use cases.
- **Goal:** Present insights for decisions using evidence-based safety metrics.

# Outline

- Business Problem
- Data
- Methods
- Results
- Conclusions

# Business Problem

The company is expanding in to new industries to diversify its portfolio. Specifically, they are interested in buying and operating aircrafts for commercial and private use, but do not know anything about the potential risks of aircraft. My task is to analyse the data and come up with three business recommendations.

# Data

The data used in this project is from the NTSB (National Transportation and Safety Board).

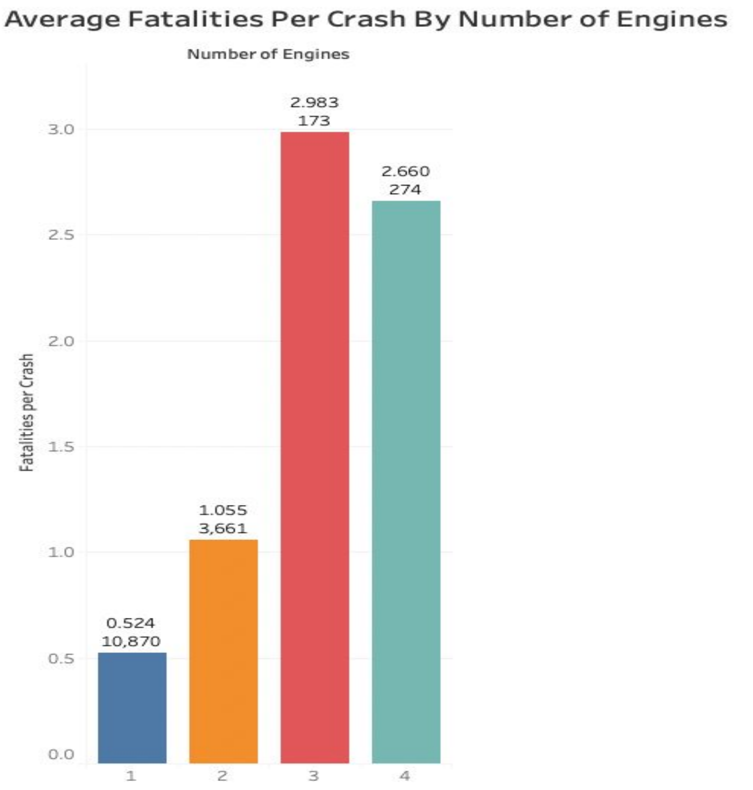
- It contains information from 1962 and later about civil aviation accidents and selected incidents within the United States.
- It contains 8889 entries (rows) and 31 variables (columns).
- Most important variables in the dataset are : Investigation type, Aircraft Damage, Category, Make, Model, Number of Engines, Engine Type, Purpose of Flight, Total Fatal Injuries, Total Serious Injuries, Total Minor Injuries, Total Uninjured, Weather Condition and Event Year.
- Limitations : It does not include data about total flight hours.

# Methods

In this project we used descriptive analysis.

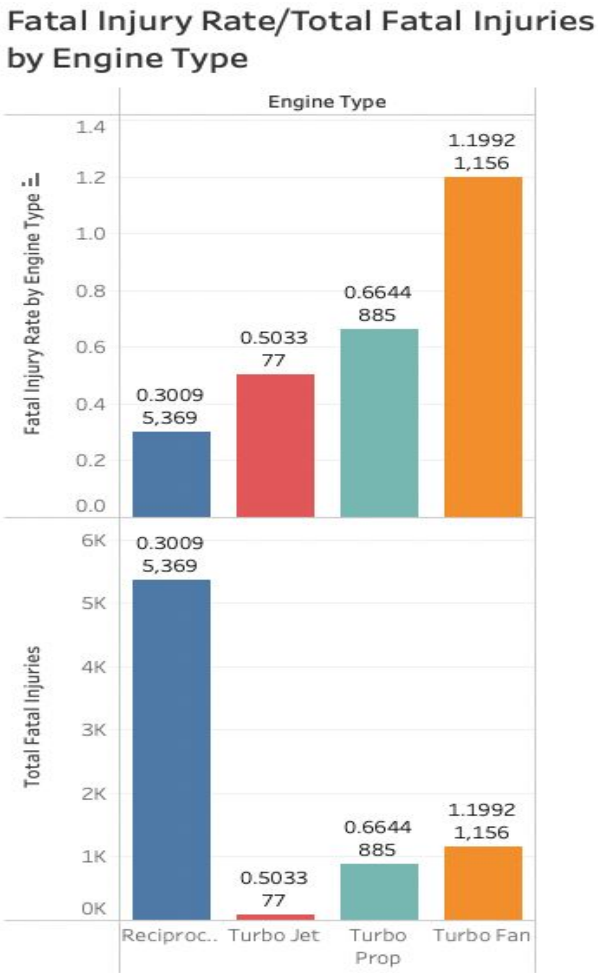
1. We understand and inspect the dataset
2. We clean and prepare the data (in this step we decide which columns we are going to use and we deal with incorrect data types and missing values)
3. Once dataset is cleaned , we perform the analysis
4. We interpret the results and create visualizations
5. Lastly , we communicate our findings

# Results



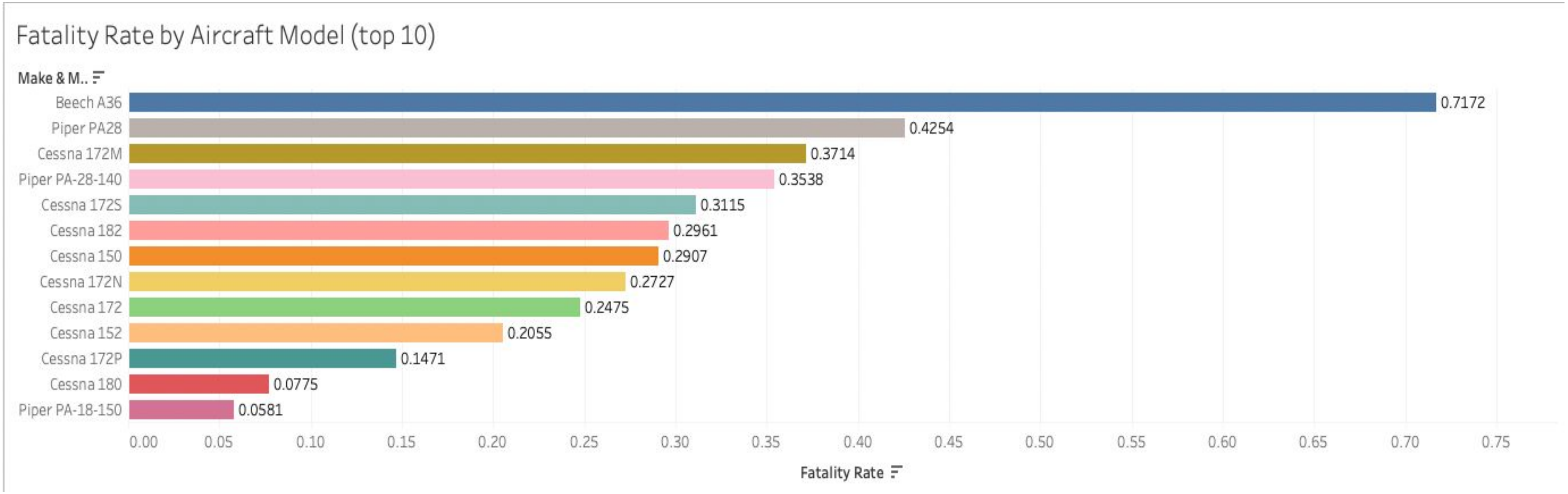

**Number of Engines:**

- 2 and 4 engines offer better redundancy and safety.
- 1 engine: lower fatality rate but highest in total fatalities.
- 3 engines: higher fatality rate per crash.




## Engine Type:

- Reciprocating: Lowest fatality rate - ideal for general aviation.
- Turbofan: Suitable for commercial operations, thought higher risk per crash.
- Turbojet: Military-focused, not recommended.




## Aircraft Models:

- Piper PA-18-150 & Cessna 180: Lowest fatality rates.
- Cessna 172 variants: Moderate to low fatality rates - safe and reliable.

# Conclusions

**Recommendations:**

1. Choose 2 or 4 engine aircraft for balanced safety and performance.
2. Use reciprocating engines for general aviation (private); turbofans for commercial; avoid turbojets.
3. Prioritize aircraft models with the lowest fatality rates:

- Piper PA-18-150
- Cessna 180
- Cessna 172 variants

**Project Limitations:** For better analysis , it would be useful to have the data of flight hours or number of flights , also for “purpose of flight” some of the variables can be refer to private or commercial , we will need more information from the company about the exact use of the aircraft to be purchased.

# Thank You!

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